Spicy Rodents — Roblox Rebuild Brief

This is the complete brief for the coding and Blender session. Build the whole game in one pass. Do not ship a walkable-town-first milestone; every role below must be test-playable from the first runnable build.

1. Source of truth

- Canonical spec: `Spicy_Rodents_v5_4_Living_Town.html`, SHA-256 `fc48b387e51ce3a454898d58d1cb3dda891b22ffad7110cdcd0ade6c777a6227`. It is live at pedagogygame.com and in the GitHub Pages repo.
- v5.4 is a clean superset of v5.2: 46 functions added, none removed. It contains the Bellwether Galleria mall (v5.3) and apple trees, bird nests, town birds, silent overflights, heavy vehicles, and fatal traffic (v5.4).
- The HTML is the reference for layout, climbing routes, traffic lanes, districts, and mechanics. The Roblox build is a from-scratch rewrite in Luau that reproduces and extends it; it does not transpile it.
- Lost in earlier HTML lineage and to be restored here: nut caches, deer, foxes, butterflies, garden insects.

2. Toolchain and repository

- Rojo project under git, one repo, Roblox Studio syncing from the filesystem. No zip handoffs.
- Blender for all character and prop assets, exported as FBX with animations, imported via the Studio animation importer.
- Code organized by concern, one module per system, under `ReplicatedStorage/Shared`, `ServerScriptService/Server`, `StarterPlayerScripts/Client`:
  - `WorldGen` — builds the town from data, never from hand-placed parts
  - `Routes` — climbing route registry (trees, poles, wires, rooftops)
  - `Traffic` — lanes, vehicles, yielding, swept collision
  - `Actors` — the single actor model (section 4)
  - `Roles` — hierarchy, queue, promotion ladder
  - `Infection` — bite, spread, cure, win condition

    - `Hospital`, `Police`, `Drone` — job systems
    - `Seasons` — in-game calendar, winter
    - `Foraging` — gardens, feeders, trash, hollows, nut meter
    - `Wildlife` — fox, deer, birds, insects
- The town is a data file: a list of lots, each with a type, footprint, facing, and lot-specific parameters. Adding a barber shop later must be a data entry, not surgery.
- `StreamingEnabled` on. The map is several thousand studs end to end at the chosen scale.
- Scale: HTML units to studs at roughly 2.5. Squirrel stands about 1.2 studs at the shoulder; humans standard R15 height.

3. The world

Districts (from v5.4, all retained)

- Old town core with streets, sidewalks, houses, utility poles and wires, trees
- Hospital (west) — expanded, see section 8
- Police station — expanded, see section 9
- Riverside old town across the river (east): stone arch bridge, cobbled quays, rooftops
- Bellwether Galleria mall (far east): six shops, upper gallery, coin fountain, interior lighting and camera constraints
- Sewer and duct interiors, house interiors

Neighborhood expansion (new)

The town should read as a real society, not a backdrop. Add, as data-driven lots:

- Restaurants: roughly double the current count. Mix of diner, pizza, taco, Chinese, café, bakery, ice cream, bar and grill, each with outdoor seating or a back-alley dumpster (foraging value).
- Barber shop, music store, hardware store, laundromat, bookshop, pet store, gas station, corner grocery, school with playground, church, park with gazebo, community garden, a small marina on the ocean side.
- Residential variety: at least eight distinct house models (bungalow, two-storey, Victorian, ranch, split-level, cottage, apartment block, large older house), and every individual house gets its own flair through paint, trim, porch, roof, and yard variants so

no two read the same. Backyards with gardens, bird feeders, sheds, fences, dog houses.
- Each lot type defines what squirrels can forage there and how risky it is.

Backdrop

The drone role sees the whole town from above, so the backdrop must hold up at altitude, not just from squirrel height.

- One side: mountains falling to rolling hills and plains, with tree line, a distant road, and fog layering.
- Other side: ocean with a beach, pier, marina, and a horizon that reads correctly from 300 studs up.
- Sky with a full day cycle: sunrise and sunset as their own graphic moments, clouds, the silent overflights from v5.4, and a genuinely beautiful night sky with stars, and a moon that runs through real phases tied to the in-game calendar (new, waxing, full, waning over a roughly 29.5-day lunar month), never a permanent full moon. A radio or cell tower in the distance with blinking red lights, placed to silhouette against the sunset.

Hollow trees

- 11 hollow trees across the town, each a claimable squirrel territory with a nut meter.
- Two premium trees have an interior pathway running up the entire center of the trunk, with chambers at several heights, and sit next to the best gardens and bird feeders.
- The other nine have a single hollow chamber reachable from a knothole.

4. The actor model (hard architectural rule)

There is no distinction in code between a player character and an NPC. There is one kind of actor per species (human, squirrel, fox, deer), and the only question is who is driving it: the AI brain or a player's controller.

- Every human and squirrel in the town is AI-driven by default.
- A player joining takes control of an existing actor mid-stride. A player leaving hands control back and the AI carries on.
- Promotion in the queue is the same handoff: release one body, possess another.

- This single mechanism implements the queue, NPC backfill, role switching, and session reset.
- The town always has more human bodies than the 35 player slots, because the hospital, police station, shops, and streets need to be staffed and walked regardless of player count.

5. Server population

- Max players 50 (Roblox default). Target population per server: 35 humans, 7 squirrels, 1 spicy rodent. Any slot not held by a player is AI.
- Squirrel population may float between 10 and 14 actors; 7 is the player-slot count, not the body count.
- Optional voluntary roles, chosen instead of a promotion: deer (several slots), fox (one slot, Easter egg; see section 11).
- Limited server count so that a real queue exists for the spicy rodent slot.

6. Roles and the promotion ladder

State this ladder in one place in code (`Roles`), and nowhere else.

1. **Human, lowest level.** Walks the town. Earns credits by working. Can be bitten.
2. **Hospital staff (5 player slots)** and **Police (5 player slots, expandable to 8).** Hospital ranks slightly above police; privileges are nearly equal.
3. **Regular squirrel (7 player slots).** Forages, stores nuts, claims a hollow.
4. **Spicy rodent (1 slot).** The zombie squirrel. The role everyone wants.

Movement between rungs:

- Base rank among humans = time logged in and playing on this server, modified by job credits.
- Squirrel rank = 90% nut count, 10% momentary stink-gland percentage. The highest-ranked squirrel becomes the next spicy rodent when the slot opens.
- When the spicy rodent dies (run over, killed by police or hospital staff, etc.), that player restarts as a lowest-level human.
- A human who kills the spicy rodent is promoted to the lowest-rank uninfected squirrel if a squirrel body is free; otherwise to the police station, or to the hospital if the rodent had infected more than 30% of the town.

- Hospital staff who kill the spicy rodent when it enters the hospital instantly become a lowest-rank uninfected squirrel.
- A regular squirrel cannot kill the spicy rodent; it can only be infected by it.
- A regular squirrel that gets infected loses heavily (points and nuts) and drops in the queue.
- Squirrels that die in winter restart as humans.
- Deer and fox are side exits a player may choose instead of a promotion; leaving them returns the player to human at their previous human rank.

7. Squirrel play

Movement

Port the v5.4 route registry: curves, sampled points, kind (branch, pole, wire, lookout, roof), speed, surface radius, and the per-route `visualOffset`. Climbing is rail-based as in the HTML, with free running on the ground and leaps between rails.

Animation (Blender)

Seven individually authored clips on a single rig:

1. Run
2. Climb
3. Wire traversal
4. Leap
5. Pounce
6. Bite dance
7. Sneak (stalk)

The squirrel must have weight and presence: fur that catches light, muscle under the skin, a tail that lags the body. This is a modelling and rigging requirement, not a lighting one. Treat the three-stacked-spheres squirrel from the HTML as a placeholder to be replaced entirely.

Foraging and territory

- Sources: gardens, bird feeders (risky — birds and homeowners), trash cans and restaurant dumpsters, apple trees, the mall plaza.

- Nuts stored in a claimed hollow raise its meter. Above 70%, the squirrel owns the tree and other squirrels cannot take its nuts.
- Premium gardens and feeders near the two big hollow trees can be raided about once per in-game week.

Stink gland

A slow-filling meter that counts for 10% of rank. It can only be fired at 95% or above, which is hard to reach, and firing it drops the meter by 30 points. The burst repels foxes and humans briefly.

8. Spicy rodent play

- Stalk (hold), pounce, bite as in v5.4. Unaware people are easier.
- Biting humans infects them; biting regular squirrels raises town-wide infection rate.
- Fatal traffic from v5.4 ports as a relative swept collision test on the server, not a Touched event. Being run over ends the spicy rodent's run.
- Hidden win path: infecting hospital staff slows cures. Critical mass of infection wins the session for the rodent. Do not surface this in the UI; let players discover it.
- Session resets when everyone is infected.

Combat dynamics

The fight between rodent and human is a real duel, not a single pounce. Build these as tunable systems:

- Timing: a pounce has a wind-up, a commit point, and a recovery; biting during the human's reaction window lands, biting into a raised arm or a kick does not.
- Approach: zig-zag movement raises the human's miss chance but costs speed; a straight sprint is fast and readable.
- Feint: a pretend lunge that forces the human to flinch or swing early, opening a real window. Feints cost stamina and, if read correctly, hand the human a free hit.
- Humans respond: flinch, swing, kick, back away, or flee, chosen by awareness and nerve. Vigilantes never flee.
- Every option that shows off should carry a trade-off, so psyching out a human is a skill, not a free win.

Vigilantes

8% of all human actors on the map at any moment have a vigilante streak. On sighting the spicy rodent they move to kick it rather than evade, and they answer the drone's call.

Health and house defenders

- The spicy rodent has a health meter. A kick does not kill; roughly two kicks without proper recovery do. Health regenerates slowly while sneaking or resting in a hollow.
- Three houses have a defender who bursts out when the rodent chases someone inside: two older housewives (different demographic appearances) with rolling pins, one man with a broom. Their blows hit 50% harder, so one blow at low health kills.

House interiors

- 40 houses have full interiors the rodent can burst into. Humans can run inside and hide; whether they escape or get bitten depends on the chase. This must actually run and be tested, not be stubbed.
- These 40 houses are also the homes a human player can claim (35 player slots, so there is slack). Each real day of play earns remodel points; remodels persist for the life of that server instance.

9. Hospital

Needs far more elaboration than the HTML. Build it as a real building: reception, triage, two wards, an isolation ward, pharmacy, staff room, ambulance bay, roof access (a squirrel route).

- Infected humans are delivered or walk in. Staff disinfect them through a short interaction loop at treatment stations.
- Hospital performance = speed of disinfection. Good performance raises staff rank and unlocks station upgrades.
- Staff can kill the spicy rodent if it enters the building (section 6 reward).
- Infected staff cure slower, feeding the hidden win path.

10. Police

- Five officer player slots at start. Officers pursue and can kill the spicy rodent.

- Good performance unlocks additional officers one at a time up to eight total; later officers are more aggressive and better coordinated.
- At eight officers, the next upgrade unlocks a drone: one player flies it. Full flight controls, camera feed, a spot-and-mark ability that reveals the spicy rodent to officers for a short window and calls nearby vigilante neighbors to the spot. Make this as good as it can be; it is the role that justifies the backdrop.

11. Wildlife and voluntary roles

- Fox: one player slot, Easter egg. Hunts regular squirrels; cannot harm the spicy rodent. Repelled by the stink gland.
- Deer: several player slots. Peaceful, wanders, can be startled by traffic and the drone. Purely for players who want it.
- AI-only: birds and nests (from v5.4), butterflies, garden insects, and restored deer and foxes when no player holds them.

12. Seasons

- In-game calendar. A week is the raid cycle for premium gardens. Time scale is one tunable constant (real seconds per in-game day); pick a sensible default and tune it in playtest. Rough targets mentioned: a winter cycle every 50 real minutes or so, with visible day and night.
- Every five in-game weeks, winter arrives and lasts one week: snow, bare trees, dimmer light, reduced foraging.
- At winter's end, squirrels whose hollow meter is below the survival threshold die and restart as humans, opening squirrel slots.

13. Credits and upgrades

- Humans earn credits by working: hospital, police, and shop jobs.
- Credits raise rank in the queue toward the squirrel and spicy rodent slots and buy the station upgrades in sections 9 and 10.

14. Deliverables

- A runnable Roblox place with every role above test-playable: human, hospital staff, police, drone, regular squirrel, spicy rodent, fox, deer.
- Blender source files and exported FBX for the squirrel (rigged, seven clips), human variants, fox, deer, and key props.
- The town data file with every lot, including the full restaurant and shop expansion.
- A short README describing the module layout and how to add a lot type, a route, or a role.